# Aavritti Soni

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## Experience

**Senior Software Engineer III** Mar 2025 – Jul 2026
*Coupang (Contract)* *Bangalore, India*

- Enhanced a shipment replanning application's JavaScript UI and backend built with Java 17 and Spring Boot to combine multiple transfer requests into one, **reducing processing time by approximately 67%**.
- Conducted non-functional resilience tests on AWS simulating traffic surges and API failures, used Grafana to track MTTR, and implemented mitigations that **cut recovery time by 33%, from 90 to 60 seconds.**
- Fixed a backlog transfer bug that returned stale Elasticsearch results, reducing repeated fetches and **cutting request response time by 2 ms**.

**Software Engineer II** Dec 2024 – Feb 2025
*Australian Securities Exchange* *Sydney, Australia*

- Optimized Elasticsearch queries and indexes to **improve read and write performance by 40%** and reduce latency while securing a Java application through dependency upgrades that preserved backward compatibility.

**Software Engineer II** Feb 2022 – Mar 2024
*Goldman Sachs* *Bangalore, India*

- Engineered Kafka workflows to ingest 10M+ records into Elasticsearch, managed deployment and hosting on AWS, and **reduced dependency on the Data Operations team by 45%**.
- Designed and deployed low-latency RESTful APIs with in-memory caching, **reducing response time by 20%** and increasing system throughput.

**Software Engineer** Jun 2019 – Feb 2022
*Bank of America* *Chennai, India*

- Contributed to the full-stack migration and rebranding of a legacy platform using C#, Angular, .NET, and MySQL, designing internal dashboards used by project managers and engineers.
- Led automation efforts by integrating Outlook, SharePoint, and Jira dashboards to streamline project testing and bug tracking, **improving efficiency by 32% and saving approximately 900 manual hours annually**. Received Bronze and Silver Awards in 2020 and 2021 for contributions to the migration and improved system efficiency.

## Technical Skills

**Languages**: Java, SQL, JavaScript, TypeScript, C++, Python
**Web Technologies and Frameworks**: HTML, CSS, Angular, Spring Boot, MVC, FastAPI, RESTful APIs
**Databases and Messaging**: MySQL, Elasticsearch, Redis, Kafka, Databricks
**Cloud and Platforms**: AWS, Linux, Docker, Kubernetes
**Development Tools**: Git, SonarQube, Maven, Jenkins, GitLab CI, CircleCI, Swagger/OpenAPI, Grafana, Gremlin

## Projects and Contributions

**Gluco: AI-Powered Wearable Health Analytics**

- Integrated Clerk authentication for secure sign-in and session verification in the FastAPI backend, supporting a team-developed wearable-data model built with scikit-learn that **achieved 60.4% accuracy**.
- Integrated an LLM-powered conversational agent using the OpenAI Agents SDK and OpenRouter to answer questions and provide insights from wearable data.

**Open Source Contributions at IOTracx**

- Smart kitchen inventory management – Designed an algorithm for the food industry to intelligently streamline supply chain workflows using machine learning.
- Automatic visual inspection of vegetables – Developed an application for analyzing surface defects on tomatoes using dlib for object detection and defect scanning at the warehouse.

## Education

**North Carolina State University, USA** May 2028
*Master of Computer Science*
*Coursework: Design and Analysis of Algorithms, Automated Learning and Data Analysis, Software Engineering*

**Vellore Institute of Technology, India** May 2019
*Bachelor of Technology in Computer Science*